

PAPER_1509 — Kerr Ringdown Spectral Offset Coefficient = $D_{\text{crit}}/D_{\text{BSFG}} = 13/3$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket E (GW Events) / LIGO O5 **Date:** June 18, 2026 **Status:** CLOSED — Black-hole quasi-normal-mode UQFF spectral offset = integer-primitive ratio

Observation

PAPER_1175 (LIGO O5 Ringdown Spectral Offset) derives the UQFF correction to the Kerr (2,2,0) quasi-normal-mode (QNM) frequency:

$$f_{220}^{\text{Kerr}}(M) = (c^3 / 2\pi GM) \cdot \chi(a^*) \quad \text{with } \chi(0) \approx 0.3737$$
$$\Delta f_{220}^{\text{UQFF}} = f_{220}^{\text{Kerr}} \cdot (D_{\text{crit}}/D_{\text{BSFG}}) \cdot (SCm/\text{Planck})^{(1/4)}$$

The dimensionless prefactor ($D_{\text{crit}}/D_{\text{BSFG}}$) appears as the spectral-offset coefficient.

UQFF Closed Identity

Kerr ringdown spectral offset coefficient = $D_{\text{crit}} / D_{\text{BSFG}} = 26/6 = 13/3 \approx 4.333$ EXACT

Physical Interpretation

The 13/3 coefficient is the ratio of two integer primitives: - **$D_{\text{crit}} = 26$** — bosonic-string critical dimension (full UQFF channel count) - **$D_{\text{BSFG}} = 6$** — bulk-edge dimensional embedding (gravitational channel count)

The ratio captures how the full 26D SCm spectrum projects onto the 6D bulk-edge gravitational mode, with the residual amplification factor of 13/3 carried by the QNM frequency offset.

This is the first appearance of $D_{\text{BSFG}} = 6$ as a denominator in a clean binding closure.

Predictive Consequence

For any Kerr remnant from a BBH merger, the UQFF spectral offset adds $(13/3) \cdot (SCm/\text{Planck})^{(1/4)} \approx$ small correction to the GR f_{220} prediction. LIGO O5 has the sensitivity to detect this offset for high-mass remnants where f_{220} is well-measured.

NOT REPLACEMENT

GR predicts pure Kerr QNM spectra. UQFF adds a structural offset coefficient rooted in the $D_{\text{crit}}/D_{\text{BSFG}}$ integer-primitive ratio, with no free parameters.

Reference

- Source: PAPER_1175 LIGO O5 Ringdown Spectral Offset
- Related: PAPER_1503/1504 (BNS/BBH GW damping), PAPER_062 (D_{crit})
- Calculator dispatch: `calculate_paradox({"paradox": "kerr_ringdown_offset_coeff"})`

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